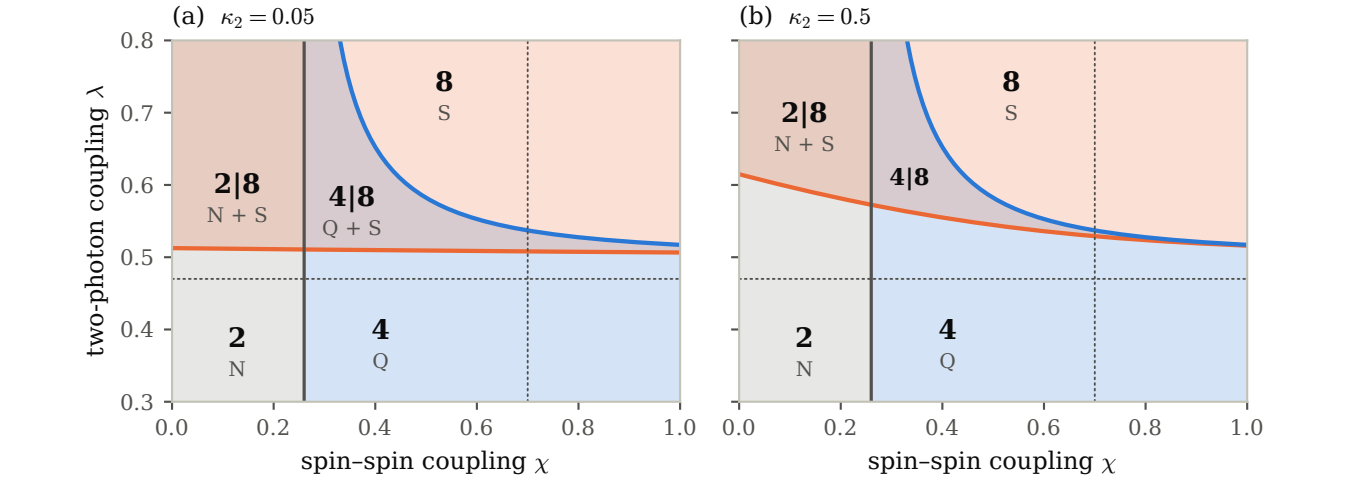

(a) $\kappa_2 = 0.05$
(b) $\kappa_2 = 0.5$
two-photon coupling λ
spin–spin coupling χ
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8
1.0
2|8
N + S
4|8
Q + S
8
S
2
N
4
Q
χ_c: spins order ($2 \to 4$, continuous)
λ_S: lobes appear (fold)
λ_p: Q unstable, $2\lambda|j_x| = \omega_c$
cuts: $\lambda = 0.47$, $\chi = 0.7$